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STATS 101C: Introduction to Statistical Models and Data Mining

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Skin Cancer Classification Model Report

Introduction

Skin cancer is the most common form of cancer in humans, mainly caused by unprotected exposure to the sun's harmful ultraviolet (UV) rays. Regular exposure to sun can cause abnormal skin cell growth, which leads to lesions on the epidermis and can cause further damage into the dermis and hypodermis and even metastasize to other organs in the body. Similarly to other forms of cancer, the scientific community is unsure of what factors other than UV exposure can contribute to skin cancer, and this paper attempts to fit a classification model on SkinCancerTrain.csv, which contains 50,000 observations of 51 variables, including one identification variable.

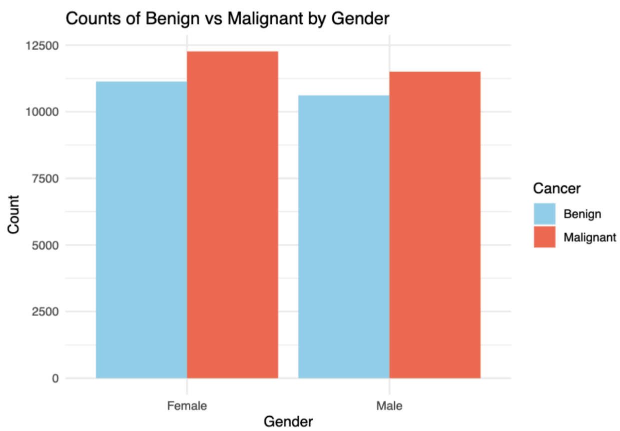
Based on background information of skin cancer, the variables that are most likely to have the largest effect on the status of a specific observation of skin cancer are those that are closely linked to UV radiation. For this reason, avg_daily_uv, sunscreen_freq, sunscreen_spf, sunscreen_type, sunscreen_brand, hat_use, and clothing_protection would seem to be the most statistically significant variables. The US Centers for Disease Control and Prevention also suggests that certain characteristics such as skin_tone and family_history may also predispose someone to be more susceptible to skin cancer[1]. Additionally, benign moles might differ physically from malignant skin cancers based on factors such as symmetry, borders, colors, size,

1. "Skin Cancer Risk Factors." CDC. 1 July 2024.
2. "Skin Cancer vs. Benign Moles: How to Tell the Difference." Griffith Dermatology. 8 April 2025.

and change[2]. For this reason, lesion_color and lesion_size might also be important to determining which factors contribute to cancer status.

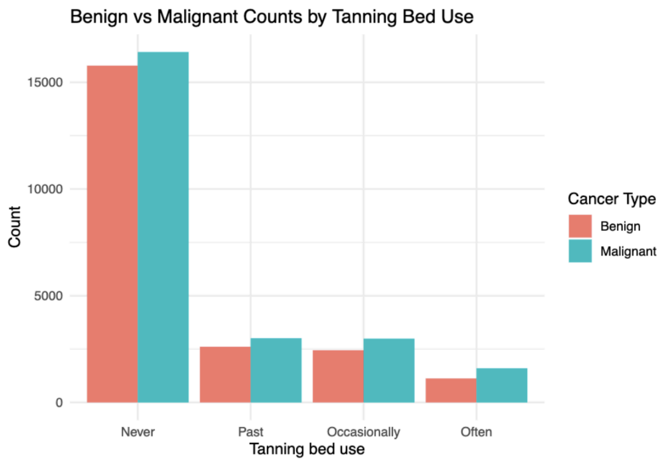
Exploratory Data Analysis

In anticipating multicollinearity, nonlinearity, and the overall relationships between the variables, we sought to perform exploratory data analysis via visualization.

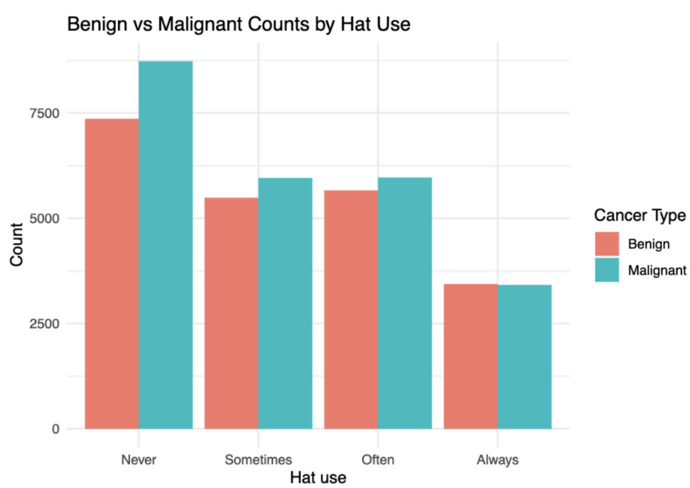


There are more observations of malignant skin cancers than benign cancers in both gender groups, however the ratio of malignant to benign in the female group is a bit larger than that in the male group.

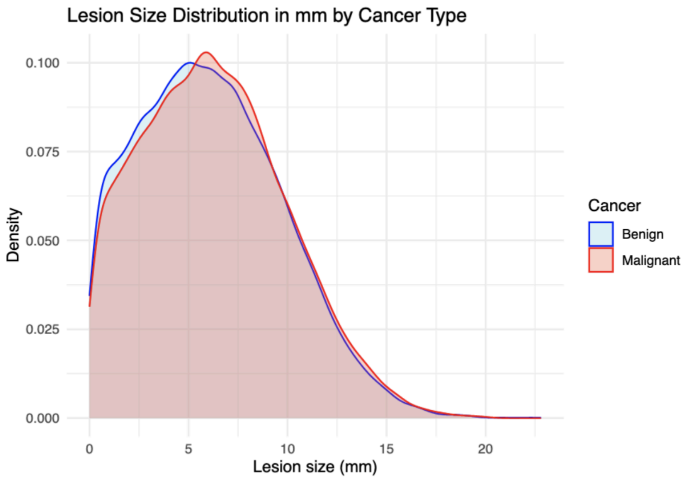
1. "Skin Cancer Risk Factors." CDC. 1 July 2024.
2. "Skin Cancer vs. Benign Moles: How to Tell the Difference." Griffith Dermatology. 8 April 2025.



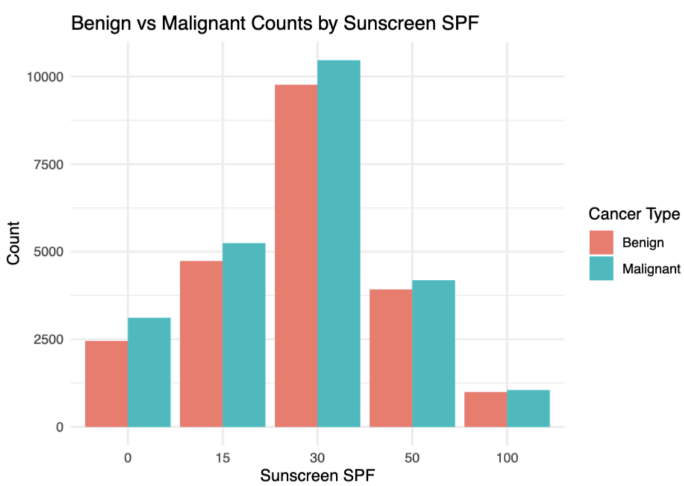
Malignant frequencies of skin cancer are higher than benign frequencies for all types of tanning bed usage. However, the proportion of malignant cancers to benign tumors cancers to be notably higher for those who use tanning beds often. It should be noted that although "often" shows the lowest number of malignant cancers, this is almost certainly because most people do not use tanning beds often, or even at all.



Ratios of benign to malignant observations are less than 1 in the never, sometimes and often categories of hat use, and steadily increases in each of those categories respectfully, while it is greater than 1 in the always category.

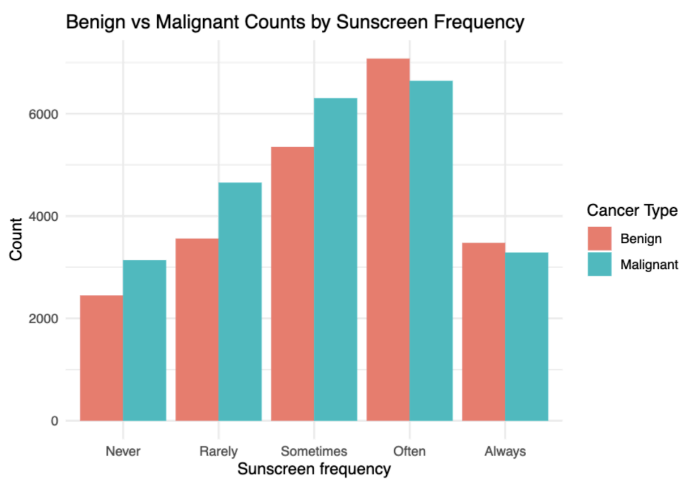


The lesion size (mm) distribution for malignant skin cancers peaks slightly to the right of the benign distribution and continues to be slightly larger than the benign distribution as the lesion size goes to infinity. The benign distribution is larger (meaning there are more observations) than the malignant distribution from size 0 to about 5 mm, which is slightly before the malignant distribution peaks.

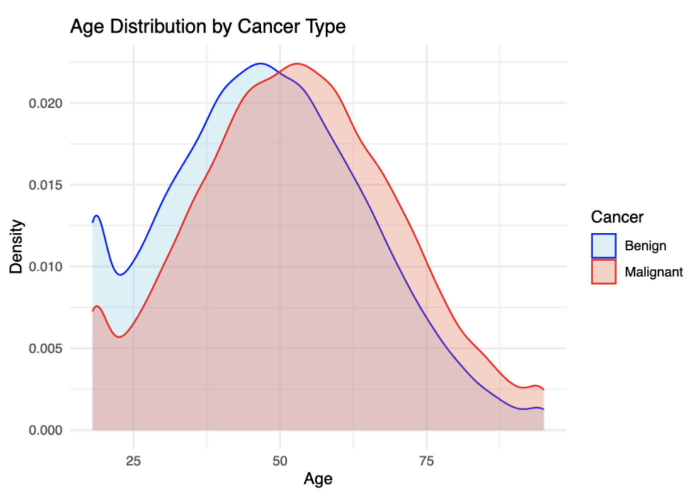


Malignant cancer frequencies are larger than benign frequencies in all sunscreen SPF groups; however the ratio of benign to malignant observations seems to be the largest in the SPFs 50 and

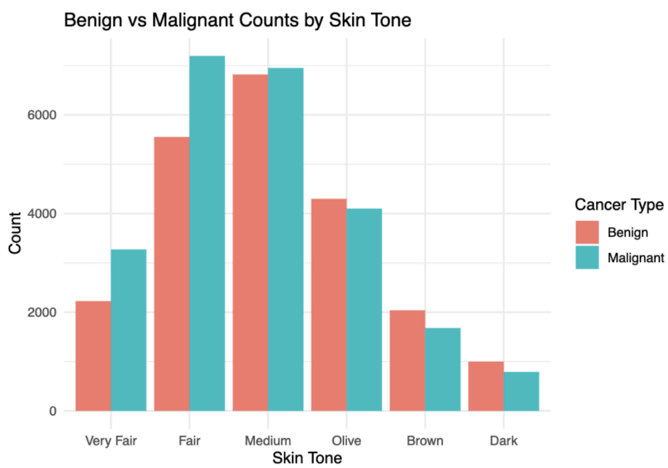
100 groups, suggesting that use of higher SPF can correlate to lower rates of skin cancer malignancy.



Benign to malignant cancer ratios are lower than 1 in the never, rarely, and sometimes categories of sunscreen frequency, while they are greater than 1 in the often and always categories. This indicates that using sunscreen more frequently correlates with lower chances of having a malignant form of skin cancer.



The age distribution of malignant cancer observations is extremely similar in shape to the benign age distribution, but it is shifted slightly to the right, meaning that age likely has some sort of positive correlation with the malignancy of skin cancer. In other words, older individuals appear to be more likely to get malignant skin cancer than benign skin cancer.



The ratio of benign to malignant in the very fair, fair, and medium skin tone groups is less than 1, while it is greater than 1 in the olive, brown, and dark skin tones. This suggests that lighter skin tones have higher rates of cancer malignancy.

Preprocessing

To prepare the dataset, we needed to ensure consistency across numeric and categorical variables, including handling missing values and encoding numeric matrices as necessary for our algorithms. Our dataset had 31 categorical variables – one being just an identifier that we removed immediately – and 20 numeric variables, so we needed to standardize them to avoid model instability, especially given the classification models we intended on using. We started by converting our target variable Cancer into a factor (with levels “Benign” and “Malignant”), and we converted our character variables to factors as well to both avoid accidental coercion into numeric form and to ensure that they were treated as categorical variables.

Our dataset had many missing values across our input matrix, so one of our primary concerns was imputing missing values. We imputed numeric variables with the mean values across the predictor, and we imputed categorical variables with the mode. Then, for models that needed numeric input, like our elastic net, principal components regression, and XGBoost model, we used one-hot encoding for respective dummy matrices. Because we had converted many of our predictor variables into factors, we also needed to modify our training and test matrices to ensure that the factor levels matched when cross-validating.

Candidate Models, Model Evaluation, and Tuning

We started with a basic GLM model trained on all of the predictors available as a benchmark to compare more complex models to. We went on to bootstrap another GLM model (resampled 50 times), perform principal component regression with elastic net regularization, and finally tune an XGBoost model.

Model	Variables used	Hyperparameters
GLM (see: glm_predictions.csv)	All predictors (raw)	N/A
Bootstrap GLM (see: submission6.csv)	All predictors (dummy-encoded)	B = 50
PCR + elastic net (see:	All components	alpha = 0.5

submission3.csv)		lambda = 0.004548542
Tuned XGBoost (see: xgb_tuned_predictions.csv)	All predictors (encoded)	nrounds = 500 eta = 0.2 max_depth = 10

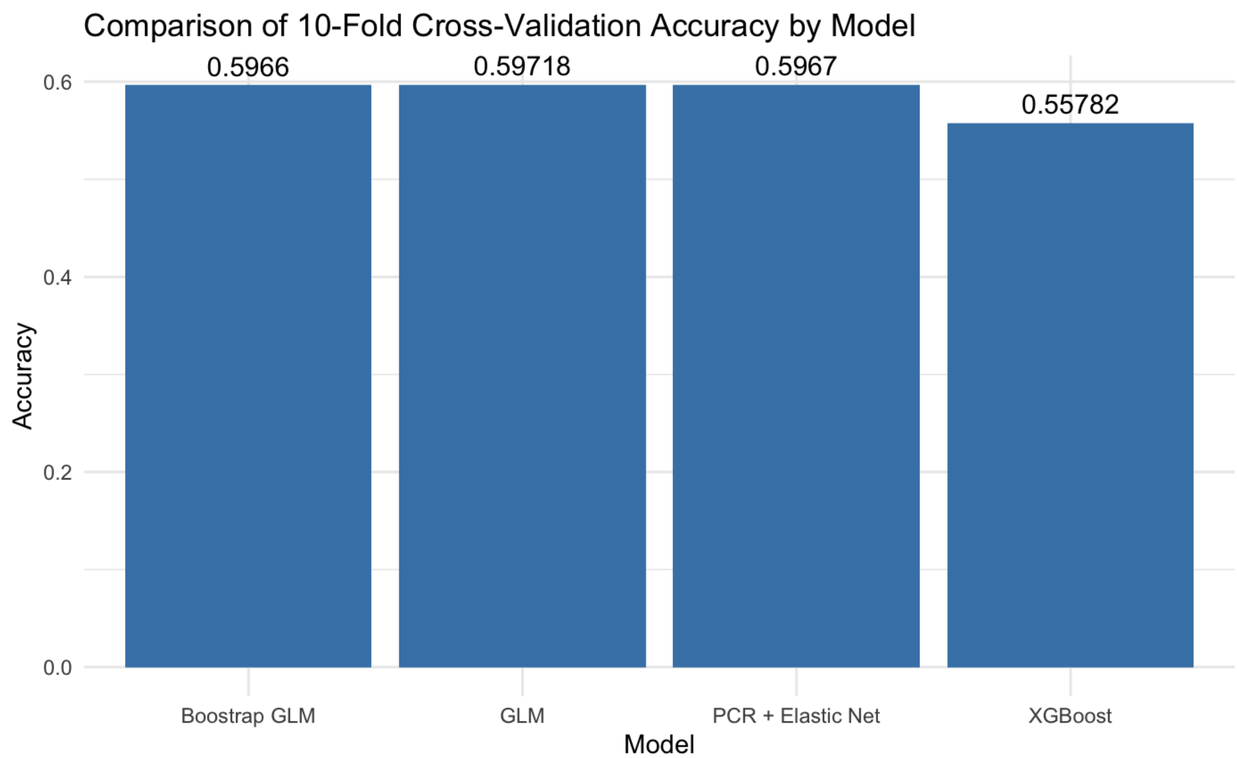
We tested the performance of each model on the half of the test data available to us on Kaggle, and they performed as follows:

Model	**Kaggle Accuracy Scores**
GLM (see: glm_predictions.csv)	0.60680
Bootstrap GLM (see: submission6.csv)	0.60460
PCR + elastic net (see: submission3.csv)	0.60670
Tuned XGBoost (see: xgb_tuned_predictions.csv)	0.60140

We tested the performance of each model on the half of the test data available to us using 10-fold Cross Validation and they performed as follows:

Model	**Performance – Mean Accuracy from 10-fold Cross Validation**

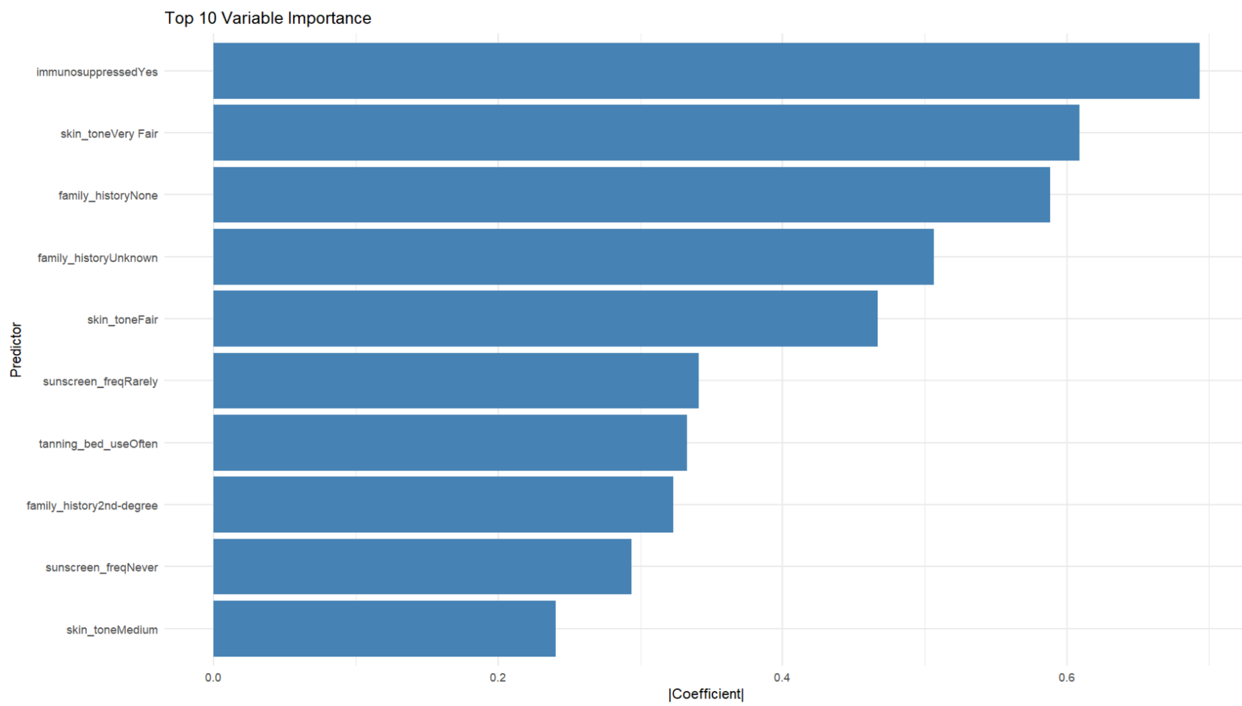
GLM (see: glm_predictions.csv)	0.59718
Bootstrap GLM (see: submission6.csv)	0.5966
PCR + elastic net (see: submission3.csv)	0.5967
Tuned XGBoost (see: xgb_tuned_predictions.csv)	0.55782



Model 1: GLM

This model was established as a baseline model prior to any resampling, cross-validating, and tuning methods, so we created a logistic regression ‘glm’ model with target variable Cancer using all predictor variables available. Our probability threshold was set at 0.5 for our binary response variable (“Malignant” or “Benign” as outcomes).

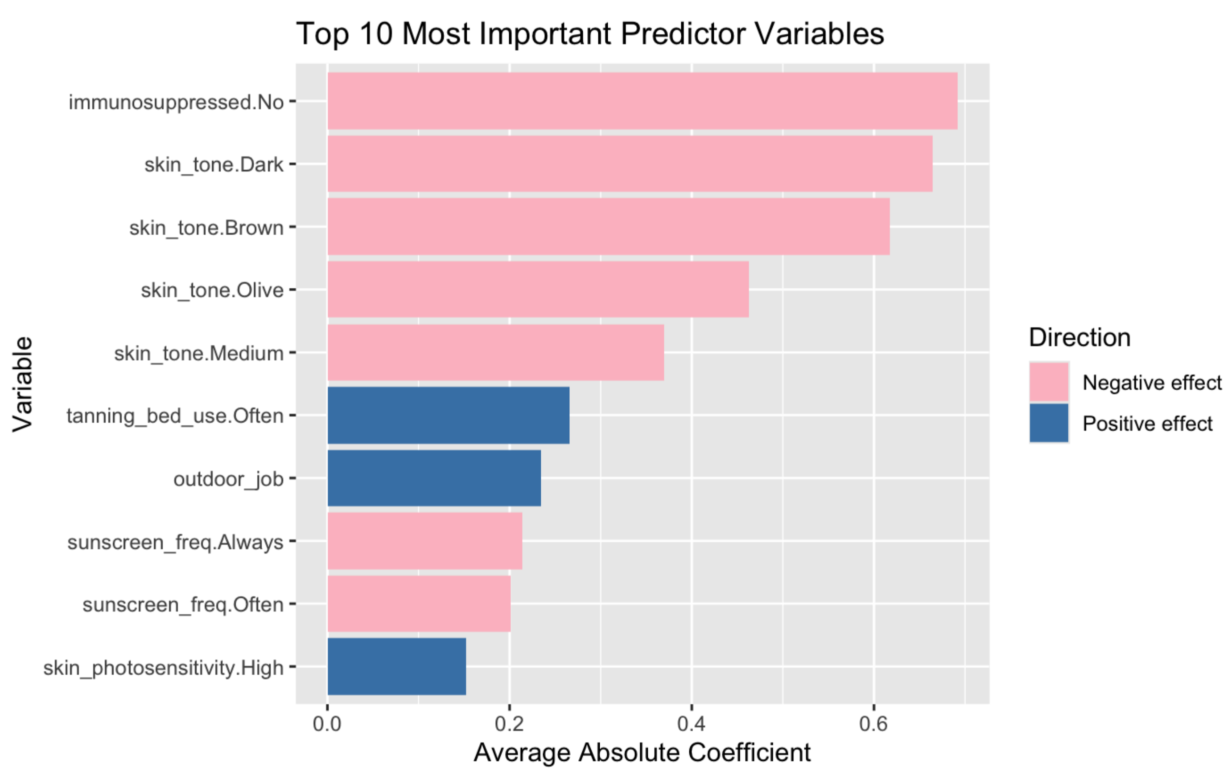
Our most important variables were immunosuppression, skin tone, and family history: particularly, status of 'Very Fair' skin and 'Fair' skin were strongly associated with Malignant outcomes. Preliminarily, lifestyle habits like sunscreen application frequency and tanning bed use were associated with Malignant outcomes as well.



A 5-fold cross-validation was performed on the GLM model in order to analyze the averaged predictive performance across the five folds. This yielded an ROC of 0.637, which suggests that the model performance is only slightly better than guessing, which would produce an ROC of 0.5. The model had a sensitivity of 0.543 and a specificity of 0.646, meaning that around 65% of benign cases are classified correctly and explaining why there is decreased sensitivity of 54% for the malignant cases.

Model 2: Bootstrap GLM

This model relies on logistic regression for binary classification, using the ‘glm’ function with the family set as “binomial”. The target variable is Cancer, same as the rest of the models, and it uses all predictor variables from the training dataset. The model differs from the standard GLM model by instead applying Bootstrap sampling to create multiple training samples from the original dataset. For each sample, a logistic regression is trained, and the predictions are averaged across all the models. This procedure helps to reduce variance and improve the stability of our predictions. In this case, 50 bootstrap samples were drawn from our training data with replacement and used to predict probabilities on the test set. The predictions were then averaged across all the models. For our final classification, we set the probability threshold to be 0.5. If probability exceeded this threshold, the Cancer class would be assigned as “Malignant”, otherwise it would be assigned “Benign”.

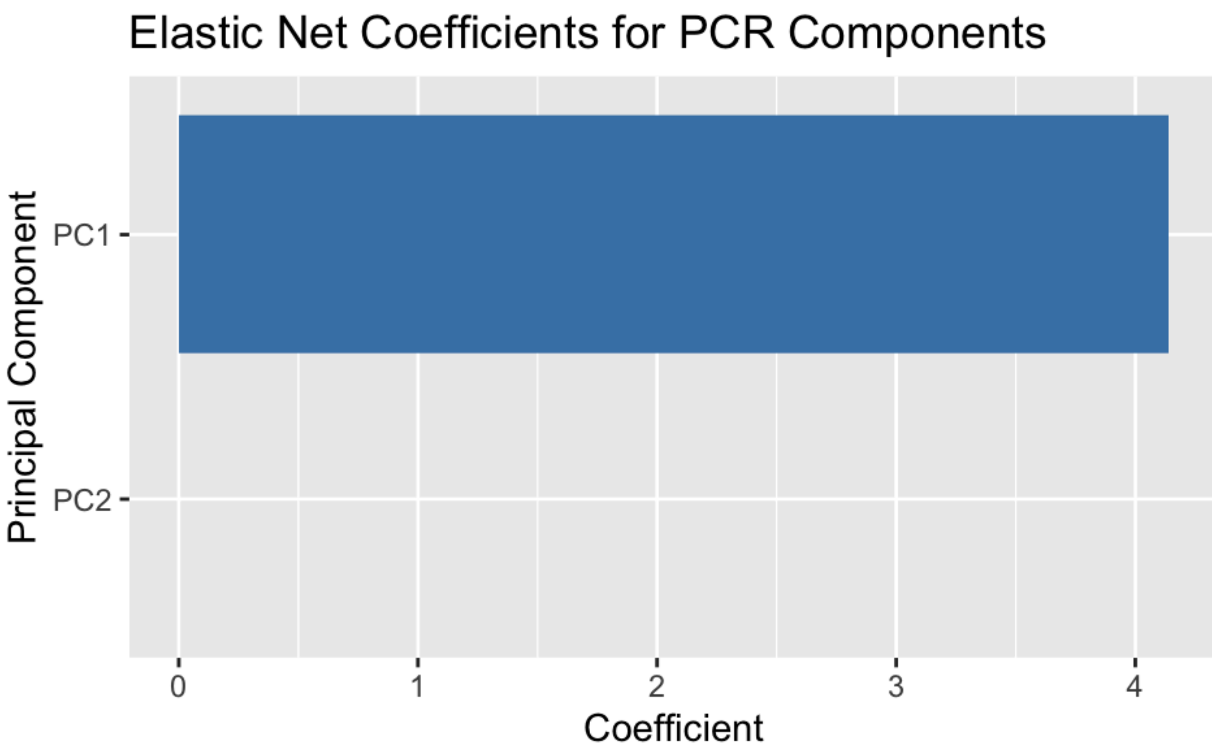


The most important features in the Bootstrapped Logistic Regression Model include whether or not the patient is immunosuppressed, skin tone, and tanning bed usage respectively. All three of these had a negative effect on the prediction, meaning that they decreased the predicted probability of Malignance, tending towards Benign Cancer. For example, if a patient was not immunosuppressed, that affected their prediction towards Benign, which was similar for dark, brown, olive, and medium skin tones. Conversely, if a patient used a tanning bed often, this had a positive effect, meaning that it pushed the prediction towards Malignant.

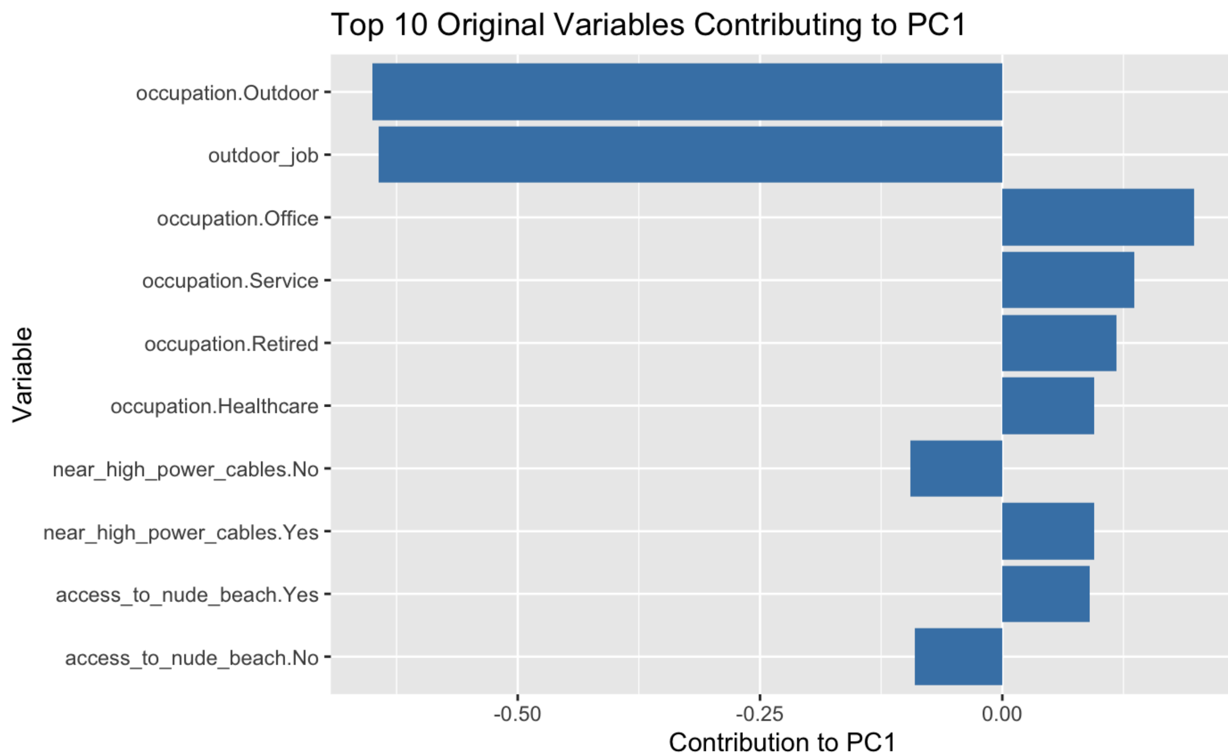
A 5-fold cross-validation was performed on the bootstrap GLM model in order to analyze the averaged predictive performance across the five folds. This yielded an ROC of 0.636, which suggests that the model performance is only slightly better than guessing, which would produce an ROC of 0.5. The model had a sensitivity of 0.543 and a specificity of 0.645, meaning that around 65% of benign cases are classified correctly and explaining why there is decreased sensitivity of 54% for the malignant cases.

Model 3: PCR + Elastic Net

This model relies on logistic regression with Elastic Net regularization for binary classification, which is carried out using “glmnet”. After the preprocessing steps, Principal Component Regression (PCR) was applied to rescue feature dimensionality and avoid collinearity. The number of principal components was selected using cross-validation and both predicted and test data were projected onto these selected components. Logistic regression was then fitted on the PCR-transformed predictors with Elastic Net. The alpha value was set to 0.5, thus combining the penalties of L1 (lasso) and L2 (ridge). The regularization strength was also selected via cross-validation in order to minimize deviance. Predicted probabilities were then computed for the test set, following the same threshold as in the Bootstrap mode: probabilities greater than 0.5 were classified as “Malignant” and everything else was classified as “Benign”.



This figure displays the coefficients assigned by the Elastic Net model to each of the principal components generated from PCR. As can be seen, only PC1 has a non-zero coefficient and we can assume that all other components were assigned zero. What this indicates is that PC1 is the only component that contributed to the prediction of cancer as Benign versus Malignant in the PCR + Elastic Net model. PC1 thus captures the largest portion of variance in the relevant predictor variables.



Knowing the PC1 drives the predictions made by our model, this figure outlines the top 10 original variables that contributed to PC1. Variables with the largest absolute loadings have the strongest influence on PC1, and therefore on the model's classification. Positive loadings indicate variables that increase PC1 and push the prediction towards Malignant, whereas negative loadings decrease PC1 and push the prediction towards Benign. For example, Occupation tended to have the strongest influence, with its different options having the 5 most dominant effects on PC1. The negative loading of the Outdoor locations specifically show that having an outdoor job tended to affect the prediction in the Benign direction. Additionally, the more indoor-based occupations of Office and Service demonstrated an effect in the Malignant Direction. More impact can also be seen based on whether the patient lives near high power cables, with the increase of Benign probability when the answer is Yes versus the opposite when the answer is No. This illustrates that the predictions made by the model reflect relationships between the patients characteristics and cancer risk, even though the predictions are being made through the principal components rather than directly from individual variables. It is important to note that these effects are measured in the context of PC1, which combines multiple features, so Occupation's influence is considered in combination with other variables rather than in isolation.

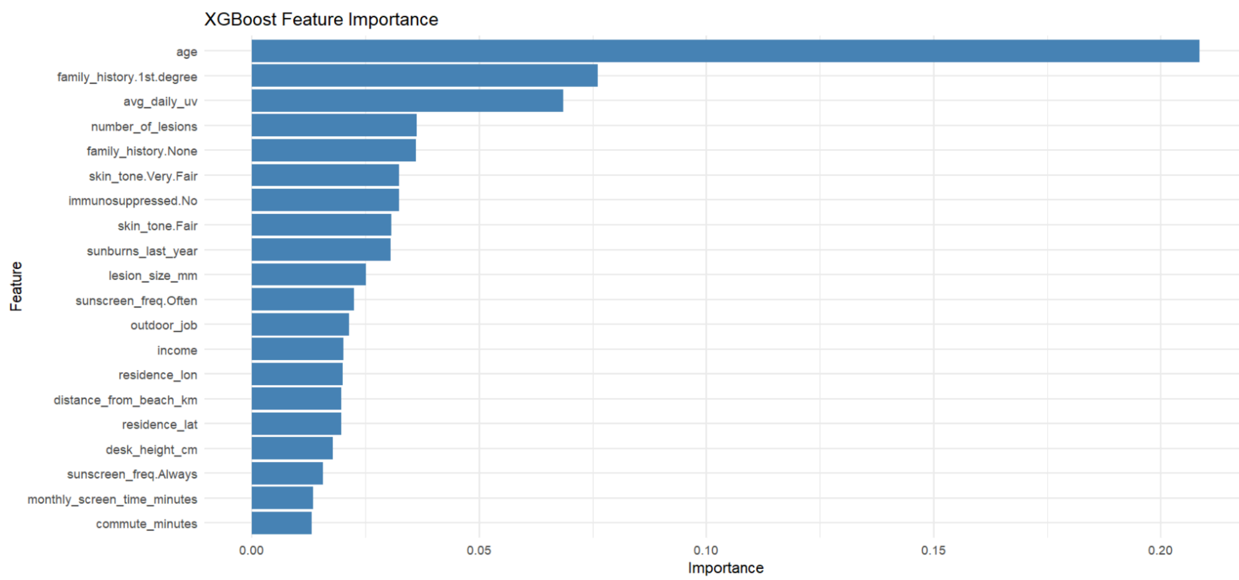
In the PCR-elastic net model, the principal components of the predictor matrix were calculated using internal cross-validation in order to discern the number of components that would optimize the cross-validation. After this, a cross-validation was completed on the elastic net section of the model in order to determine a penalty level to minimize binomial deviance. The range of deviances was concluded to be (1.384124, 1.320312), achieved by a lambda of 0.004994415. Since these deviance levels were relatively high, we could conclude that the model lacks substantial predictive capabilities. The internal structure of this model is stable, but the overall predictive performance of the model is weak.

Model 4: Tuned XGBoost

We created a gradient-boosted model to more quickly predict the classification outcomes, and added some default hyperparameters to tune the model. In using the residuals from previous classification trees to improve on the current model, we aimed to create a model that held onto accuracy in predictions but was more efficient.

Our cross-validated accuracy of this model was not particularly better than the others: to improve on this model, especially via hyperparameter optimization, we would perform further grid search and random search tuning to ensure that the boosted model is an improvement from the others with fewer hyperparameters.

Interestingly, the more impactful predictor variables for our XGBoost model were very different from the others: age had the strongest association, and other highlighted features like family history and UV exposure, while important, were significantly less important. That being said, we can also interpret the strong relationship with age as a result of detection, diagnosis, and attention later in life, and in evaluating the relationship between the variables, we keep in mind that predictors that show up consistently in all of our models may be the most practically impactful and interpretable.



A 5-fold cross-validation model was performed on the tuned XGBoost classifier after categorical predictors were encoded. The produced metrics were an AUC of 0.5791542, accuracy of 0.55782, sensitivity of 0.5944769, and specificity of 0.5176671, which suggests that when predicting a benign or malignant status for a specific observation, the XGBoost model only has the power to predict slightly more than guessing randomly.

Discussion of Final Model

Four modeling approaches were evaluated through the course of this project, including a bootstrapped logistic regression, a PCR and elastic net hybrid model, and a tuned XGBoost pipeline. Although these alternative models offered various strengths, such as variance and dimensionality reduction, our final model choice was to use standard logistic regression.

This decision was guided by two key factors. Firstly, in the Kaggle competition associated with this project, our standard logistic regression model achieved our highest prediction accuracy score among all the models we tested. The differences in performance from all of our submission was very small, with only a 0.0054 distinction from the highest-scoring to the lowest-scoring, and we are aware that it could have been from random chance that standard GLM scored the highest, however when 10-fold cross validation for accuracy was carried out on all the models, the standard logistic regression model performed the best and had the highest score.

Additionally, logistic regression is favorable because of its interpretability and simplicity. GLM provides clear, interpretable coefficients that indicate the direction and magnitude of each predictor's effect on malignancy. This is a major advantage over XGBoost and PCR-based models which tend to transform or combine predictors in ways that obscure their direct, individual impact – while the focus on interaction between variables is beneficial for inference, we aimed for more straightforward prediction. Because this project is being performed in a medical context where clarity in factors is important, logistic regression was the best choice.

Despite this praise, there is some reason to be cautious about this mode. Logistic regression assumes that there are linear relationships between the different predictors and the log-odds of the Cancer variable, which may tend to oversimplify occurrences: though can minimize variance this way, we risk increasing the bias. Relatedly, model interactions and nonlinear patterns remain unmodeled unless they are explicitly added. The imputations we made in our preprocessing may also affect subtle patterns in the data that could be otherwise adapted for with a more flexible modeling approach – for example, opting to impute with the mean instead of the median assumes a degree of centeredness throughout the input matrix that may impact our accuracy if there is significant skew.

Overall, there are several ways the predictive performance and interpretability of our model could be improved. For instance, nonlinear or interaction terms could be added to the GLM in order to capture more complex medical phenomena that the simple model may miss. The more flexible approaches, such as XGBoost could also be revisited with more extensive hyperparameter tuning, as their predictive performance is highly sensitive to parameter choices. From a data standpoint, further clinical information could likely increase predictive accuracy. This could include dermatological assessments or medical history, as much of the provided data is demographic-based. Finally, more sophisticated imputation techniques may help to preserve the natural variability in the data and reduce the bias introduced by mean or mode replacement.

Appendix

Alice:

Carried out preprocessing and shared code with other group members. Contributed to coding the candidate models that were submitted to Kaggle. Provided descriptions, figures, and explanations of model 2 & 3 in the paper. Conducted tenfold cross validation on models 1, 2, & 3 to determine their accuracy scores. Provided the explanation of the final model, its strengths and weaknesses, and improvement that could be made on the project. Communicated in the group chat with other group members.

Valérie:

Wrote preprocessing, descriptions, explanations for models 1 & 4 in the paper. Contributed to coding the candidate models submitted to Kaggle. Contributed to model evaluation, explanation of strengths and weaknesses, and group communication.

Zane:

Completed the introduction and performed exploratory data analysis on the data. Completed the cross-validations for each of the 4 submitted models and described their processes and outcomes.

Eden and Elly:

Eden and Elly both focused on the regression project. Also helped brainstorm ideas at our original group meeting for cleaning data, candidate models, and report formatting, and continued to communicate in the group chat. Ran code on their own devices to ensure reproducibility. Read through and helped edit the report.